ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 14 March	Mid cycle.
3	**(b)** the one with an eye grows	Bud needed.
4	**(c)** wings or hairs	Light structures.
5	**(b)** oviduct	Tube.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** 13 to 15 March	Around ovulation.
8	**(c)** cambia of stock and scion are in contact	Tissue contact.
9	**(a)** Both A and R are true and R is the correct explanation of A.	With one parent and no gamete fusion, the offspring are clones of the parent.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true but describe different organisms and methods.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1).	1+1
12	Pollen grains carry male gametes (1); Ovule has the egg; becomes seed after fertilisation (1).	1+1
13	IVF (test-tube baby) (1); Artificial insemination (1).	1+1
14	Buds (eyes) grow using stored food (1); Vegetative reproduction of a tuber (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1).	1+1+1
16	Seeds from sexual reproduction give variation (1); Cutting is a clone (1); Mitosis vs meiosis and fertilisation (1).	1+1+1
17	Unisexual flowers (1); Anthers and stigma mature at different times (1); Different positions (1).	1+1+1
	OR: Frog: thousands of eggs, no care (1); Hen: few eggs, some care (1); Human: one baby, extensive care (1).	1+1+1
18	Retains the quality of fruit (1); Faster bearing of fruit (1); Combines good stock and scion traits (1).	1+1+1
	OR: Poor nutrition and education (1); Pressure on resources (1); Unemployment (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Binary fission (1). (ii) Budding (1). (iii) By spores formed in sporangia (1). (iv) Number doubles every hour: $2^4 = 16$ (2).	1+1+1+2
20	Testes, ovaries (1); Oviduct and fertilisation (1); Implantation (1); Placenta and cord (1); Birth (1).	1+1+1+1+1
	OR: Nutrition (1); STIs and prevention (1); Substance abuse effects (1); Hygiene (1); Awareness and medical care (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.
- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** chance of fertilisation	Fertility.
3	**(d)** twig of the desired plant	Grafted part.
4	**(b)** water	Floating husk.
5	**(d)** epididymis	Mature.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** 1/2	Independent events.
8	**(d)** the root system	Base.
9	**(d)** A is false but R is true.	Transfer to a flower of a different plant is cross-pollination, so A is false; R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	They share the same chromosomes, including the sex chromosomes, so the sex is the same.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Different eggs and sperms (1); Different genes (1).	1+1
12	Individual can live without reproducing (1); Species ends if none reproduce (1).	1+1
13	Brings variation (1); Helps in adaptation (1).	1+1
14	Lack of oxygen (1); Seed needs air for respiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Seeds = 50 (1); Seedlings = 20 (1); 40 per cent of 50 (1).	1+1+1
16	Lining retained by progesterone (1); Uterus enlarges (1); Oxytocin causes contractions (1).	1+1+1
17	Sex-selective abortion is illegal (1); Equality of girls and boys (1); Counselling and law (PCPNDT) (1).	1+1+1
	OR: Coconut: water (1); Maple and cotton: wind (1); Xanthium: animals; balsam: explosive (1).	1+1+1
18	Bisexual: both stamen and carpel, e.g. hibiscus, mustard (1); Unisexual: only one, e.g. papaya, cucumber (1); Pollination differences (1).	1+1+1
	OR: Hen: internal; sperm enters the female (1); Frog: external; sperm released in water (1); Number of eggs differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Male: stamen (anther and filament); female: pistil (stigma, style, ovary) (1). (ii) Transfer of pollen grains from the anther to the stigma (1). (iii) Vegetative propagation by stem cutting (1). (iv) The new plant is identical to the parent and bears flowers or fruits earlier / plants without seeds can be grown (1+1).	1+1+1+2
20	Seeds are from sexual reproduction with variation (1); Genes recombine (1); Use grafting or budding (1); Steps of grafting (1); Clones keep traits (1).	1+1+1+1+1
	OR: Plant: pollen and ovule, pollination, fertilisation inside the ovule (1); Frog: sperm and egg meet in water (1); Plant: seed and seedling; frog: tadpole (1); Plants disperse seeds, frogs move as adults (1); Both show alternation of generations of stages (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that budding is the method of *Amoeba*; *Amoeba* divides by binary fission, whereas yeast and *Hydra* reproduce by budding.
- Confusing pollination (transfer of pollen) with fertilisation (fusion of male and female gametes).
- Saying that fertilisation takes place in the uterus; it takes place in the fallopian tube, and the embryo implants in the uterus.

- Writing that offspring of asexual reproduction show variation; they are identical to the parent (variation arises in sexual reproduction).
- Giving 46 as the number of chromosomes in a human sperm or egg; gametes have 23, the zygote has 46.
- Thinking the sex of the baby is decided by the mother; every egg carries X, so the type of sperm (X or Y) decides.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 1 : 1	Equal X and Y sperms.
3	**(a)** branch while it is still attached to the parent	Rooting on the stem.
4	**(a)** low water content and food stored	Dormancy.
5	**(b)** progesterone	After ovulation.
6	**(c)** A is true but R is false.	A is true; R is false because each spore has a low chance.
7	**(b)** girl	XX.
8	**(b)** banana	Seedless.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A large number of eggs compensates for the low survival chance.
10	**(c)** A is true but R is false.	A is true, but sperm carry either X or Y (egg always X), so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sperms swim to the egg (1); No pollen tubes (1).	1+1
12	Sexual: passed on with variation (1); Asexual: the clone carries the same (1).	1+1
13	No hCG from embryo (1); Progesterone falls and menstruation occurs (1).	1+1
14	Dispersed by air (1); Reach new food sources (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mitosis: body cells, 2 cells (1); Meiosis: gonads, 4 cells (1); Chromosome number maintained vs halved (1).	1+1+1
16	Planaria: whole body from a piece (1); Humans: limited, tissue repair (1); Cells divide and differentiate (1).	1+1+1
17	Dicot seed (1); Radicle to root; plumule to shoot (1); Cotyledons give food; seed coat protects (1).	1+1+1
	OR: Sperm: tail for swimming, mitochondria (1); WBC: engulf germs (1); Nerve: carries impulses (1).	1+1+1
18	Gametes have half the number (1); Fertilisation restores it (1); Meiosis and fertilisation (1).	1+1+1
	OR: Protects seeds and helps dispersal (1); Food for humans (1); Source of seeds and economic crops (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Fertilisation (1). (ii) In the fallopian tube (oviduct) (1). (iii) It supplies food and oxygen to the embryo and removes its wastes (1). (iv) $2^3 = 8$ cells (1); the zygote has 46 chromosomes (23 from the sperm and 23 from the egg) (1).	1+1+1+2
20	Days 1 to 5: lining is shed (menstruation) (1); About day 14: an egg is released from the ovary (1); Lining thickens to receive the embryo (1); Fertilisation: embryo implants, no menstruation (1); No fertilisation: lining is shed and the cycle repeats (1).	1+1+1+1+1
	OR: Select flowers and a time (1); Count visitors (1); Record in a table (1); Analyse the most common pollinators (1); Conservation: gardens, reduce pesticides (1).	1+1+1+1+1

Mistakes to avoid:

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